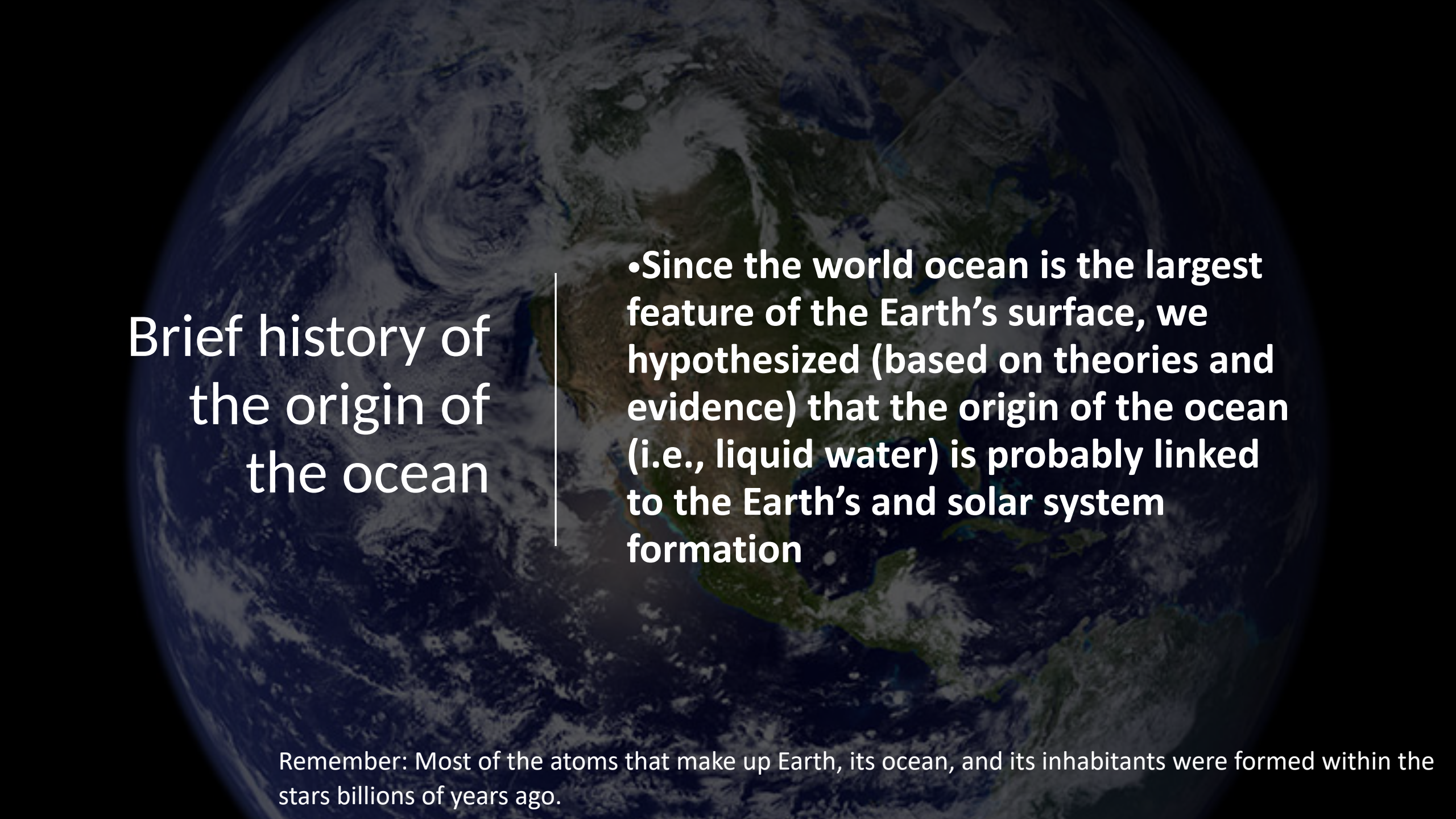




From lecture session 2: We gained some historical perspective about ocean exploration, learned how oceanography emerged as a science, and how to locate ourselves on the Earth's surface.

Lecture Session 3

1. The origin of the Solar System and the Earth;
2. Source of water that fill our ocean;
3. Connecting life with the oceans

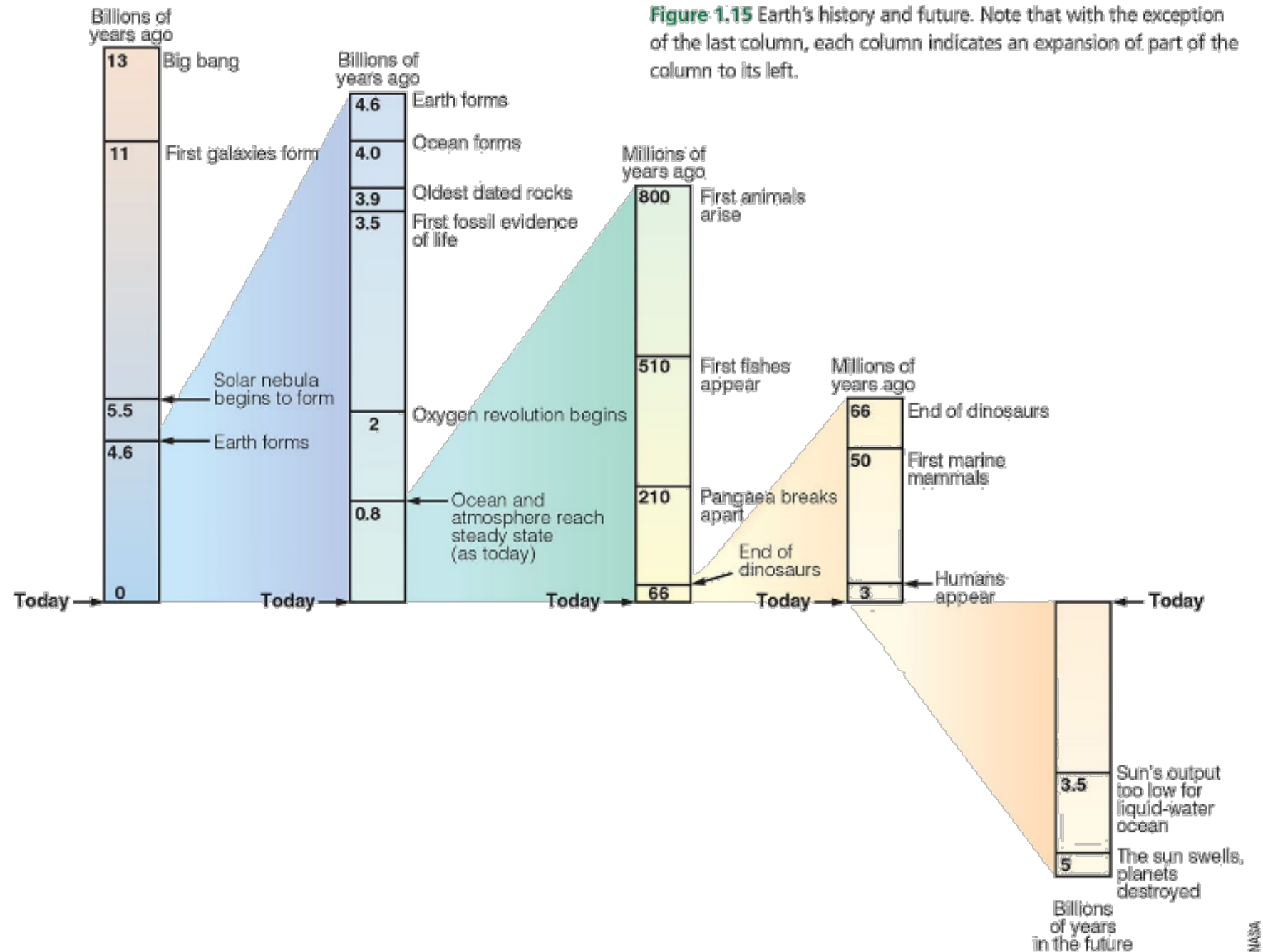


Brief history of the origin of the ocean

- Since the world ocean is the largest feature of the Earth's surface, we hypothesized (based on theories and evidence) that the origin of the ocean (i.e., liquid water) is probably linked to the Earth's and solar system formation

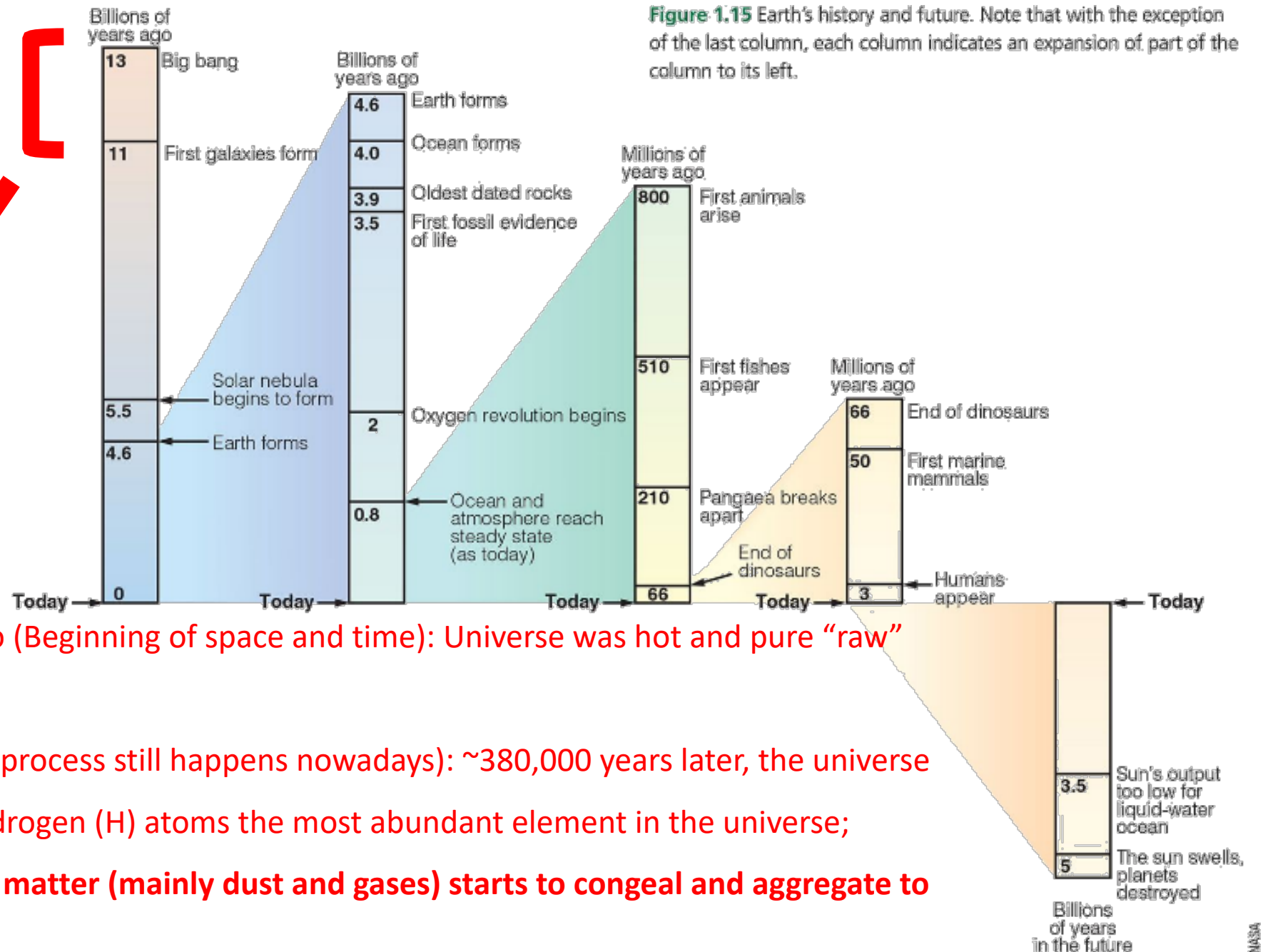
Remember: Most of the atoms that make up Earth, its ocean, and its inhabitants were formed within the stars billions of years ago.

Universe and Solar System Formation



Universe and Solar System Formation

The beginning of everything

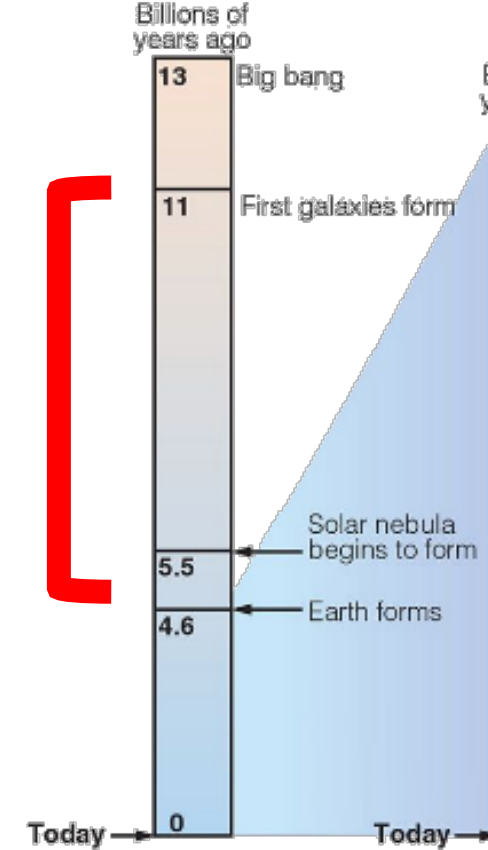


1. Big Bang ~13 Billions of years ago (Beginning of space and time): Universe was hot and pure “raw” energy;
2. Universe expansion and cooling (process still happens nowadays): ~380,000 years later, the universe became cool enough to form Hydrogen (H) atoms the most abundant element in the universe;
3. In the course of ~1 billion years, matter (mainly dust and gases) starts to congeal and aggregate to form the first stars and galaxies

Universe and Solar System Formation

From heavier elements to Earth: Stars produced all matter we find today (and continue to do so)

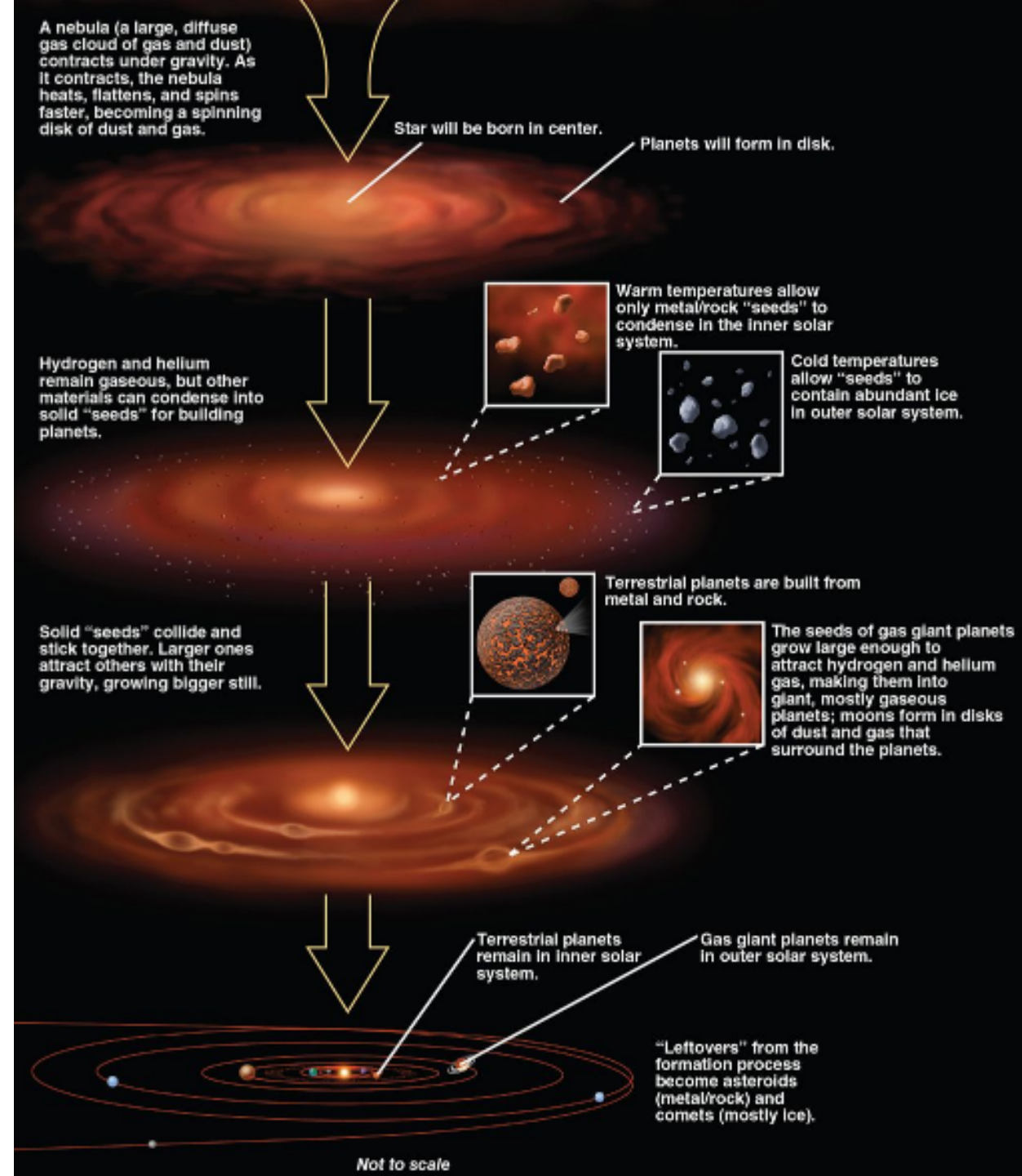
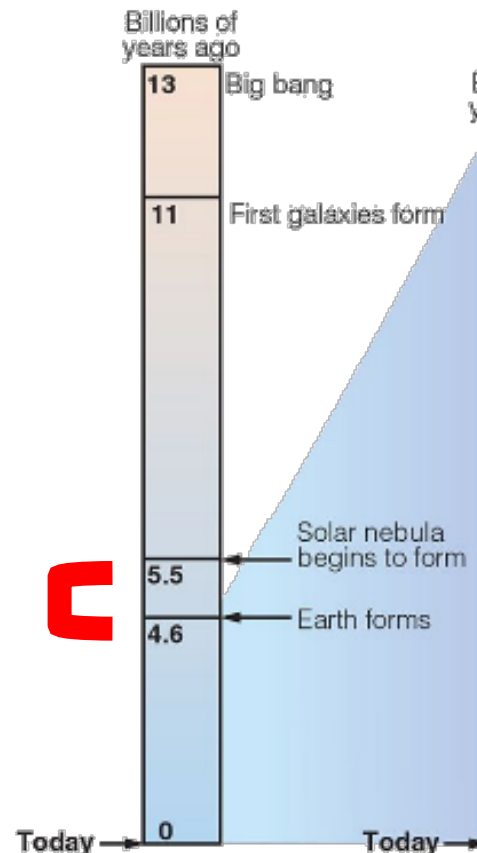
- Nuclear Fusion inside stars: Mainly produces Helium (He), however stars can produce elements as heavy as Carbon (C);
- Supernovas and the death of stars: by the final stages of the life cycle of stars they explode forming supernovas. During such explosive events, the large amount of energy liberated are partially used to create even heavier elements like Iron (Fe) and Nickel (Ni), **forming the building blocks of more complex molecules (water in the universe probably originated during this period).**



Our Sun and Earth formed from a dying star via the process of **accretion** ➡

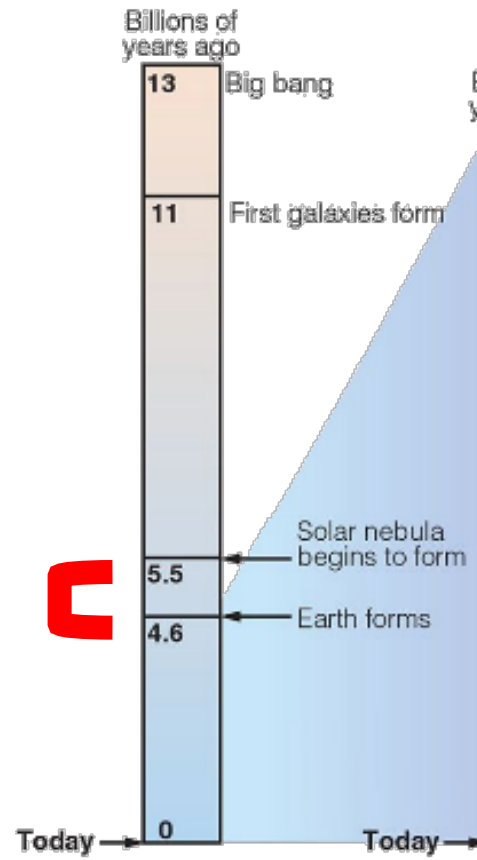
The accretion process that formed our Solar System lasted ~30-50 millions of years

Key Concept: Density = mass/volume

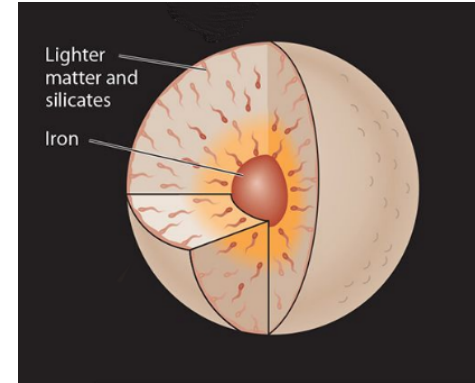
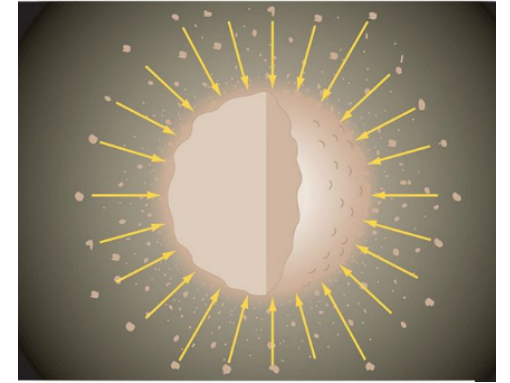


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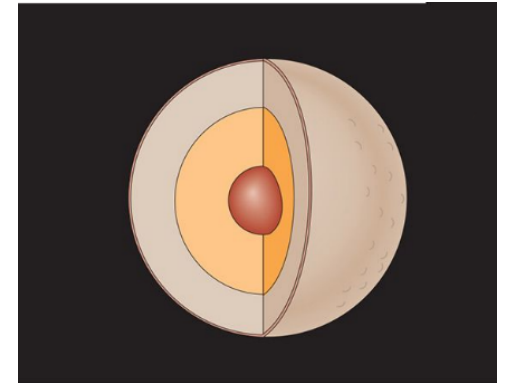


The planet grew by the aggregation of particles. Meteors and asteroids bombarded the surface, heating the new planet and adding to its growing mass. At the time, Earth was composed of homogenous mixtures of materials.



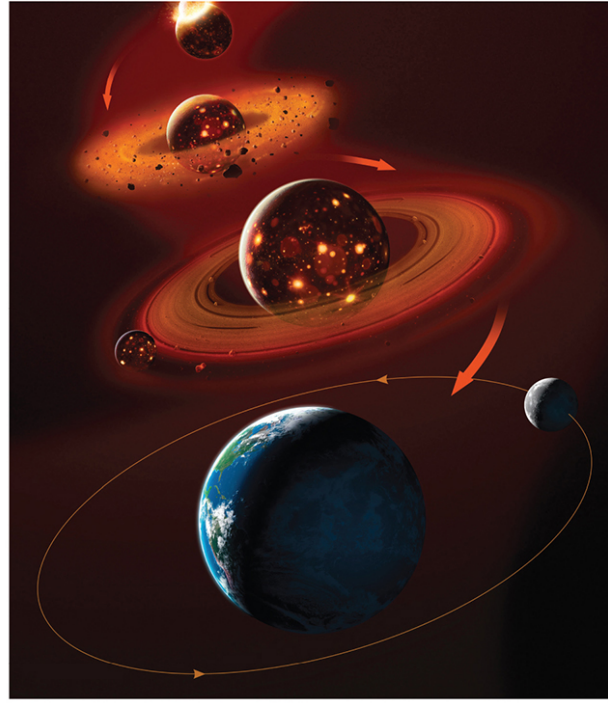
High temperatures in the interior turned the inner Earth into a semisolid mass; dense iron (red drops) fell toward the center to form the core, while less dense silicates moved outward. (**a.k.a. Planetary Differentiation**).

The result of density stratification is evident in the formation of the inner and outer core, the mantle, and the crust.





NASA/JPL-Caltech



MARK GARLICK / Science Source

Note: In the middle of the turmoil of Earth's early days our moon was formed

a The first stage of the formation of the moon. A planetary body somewhat larger than Mars is thought to have smashed into the young Earth about 4.4 billion years ago.

b The rocky mantle of the impactor was ejected to form a ring of debris around Earth, and its metallic core fell into Earth's core and joined with it. This figure illustrates the stages of the moon's formation.

c Rocks brought from the lunar surface by Apollo astronauts suggest the ejected material condensed soon after to become our moon. This photo was taken from the Apollo 11 spacecraft on its journey home.

Figure 1.10 The formation of the moon.

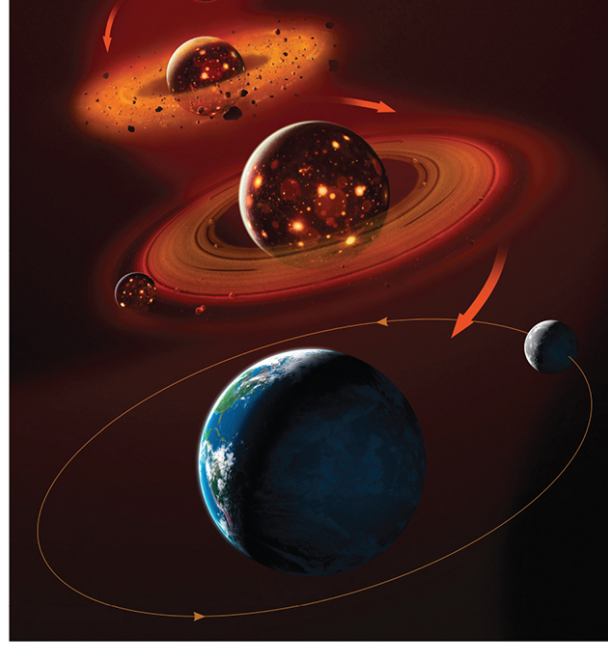


NASA



NASA/JPL-Caltech

- a** The first stage of the formation of the moon. A planetary body somewhat larger than Mars is thought to have smashed into the young Earth about 4.4 billion years ago.



MARK GARLICK / Science Source

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Figure 1.10 The formation of the moon.



NASA

Note: In the middle of the turmoil of Earth's early days our moon was formed

We will see later in the course that the moon not only control our tides, but also have evolutionary importance for some species, and influence a wide range of ocean motions (Ex: heat and nutrients distribution in places like the Gulf of Maine).

Thinking moment

A differentiation-like process is a general principles where a system separates into distinct layers or components based on different properties. Just as a planet separates into a dense core and a lighter crust, similar mechanisms based on density, size, or function can be observed in many other contexts, resulting in a stratified distribution of matter.

Which of the following processes does NOT result from a differentiation-like mechanism?

- ☐ A separation of the ocean into distinct density layers
- ☐ The formation of different tissue types (e.g., skin, muscle, bone) from a single fertilized egg during embryonic development
- ☐ The conversion of light energy into chemical energy to create organic matter by plants
- ☐ The settling of different-sized particles of sand, silt, and clay into horizontal layers at the bottom of a lake.

4.6 Billions of year ago Earth finally formed (our ocean has a home)

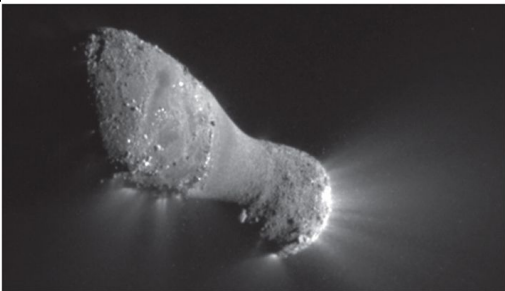
1. At first, Earth was very hot: Only gases (H_2O , CH_4 , NH_3 , H_2 , CO) and molten rock;

2. Cooling and the origin of liquid water: Rocks at the surface started to form a solid crust;

3. Water vapor trapped under the crust was released to the atmosphere via volcanic activity (outgassing). Constant water condensation helped cooling the Earth's surface and liquid accumulated over the crust. **This accumulation originated large bodies of water such as the ocean.**

4. Liquid water at the surface (and life) only exist on Earth because our especial position in the Solar System (a.k.a. Habitable Zone).

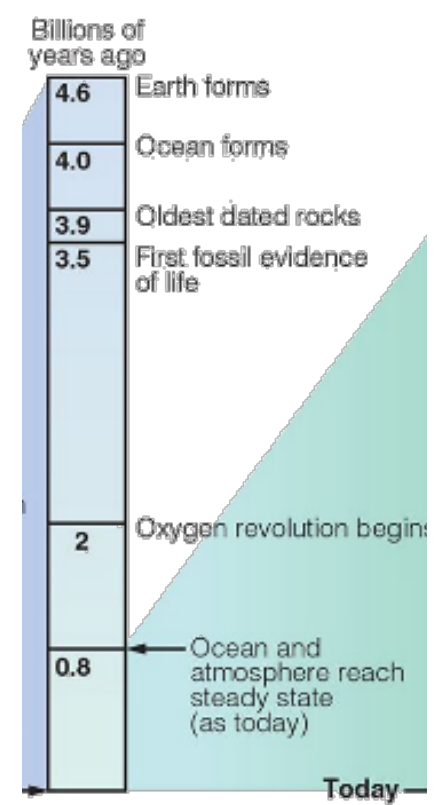
5. Asteroids and comets were external water sources to Earth



Asteroid

NASA/JPL-Caltech/UMD

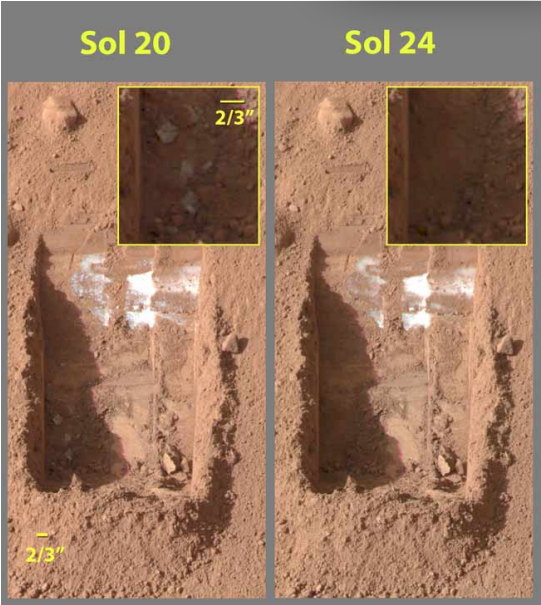
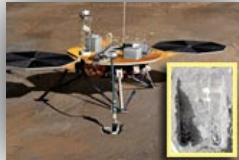
Ocean forms around 4.0
Billions of years ago



Tom Garrison

Outgassing

Remember that water is not an uncommon substance in the universe



Just a few examples

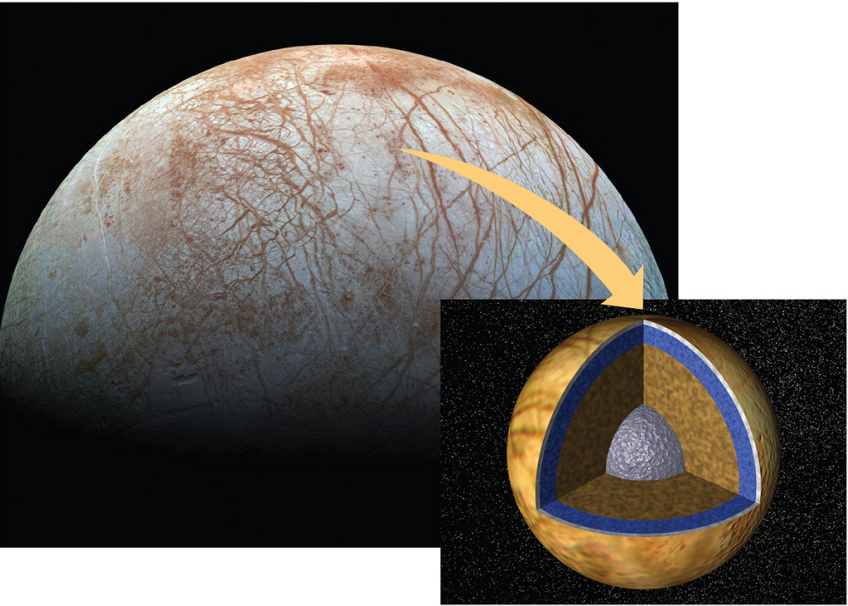


Figure 1.16 The icy, fractured crust of Europa shrouds a potential salty ocean of ice (white cross section) and liquid water (blue cross section) hidden beneath.

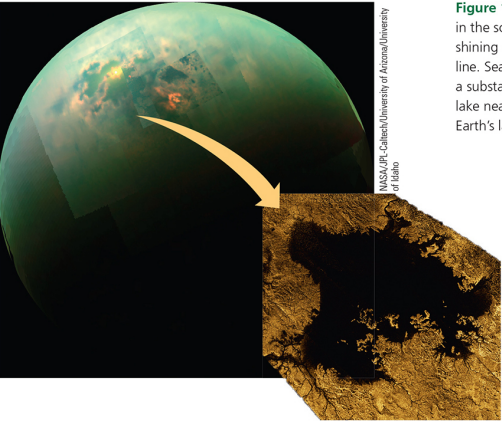


Figure 1.17 The face of Saturn's moon, Titan, the only other body in the solar system known to possess liquid on its surface. Sunlight is shining on the edge of a huge circular storm just below the day/night line. Seas of liquid methane or ethane, seen as dark patches, cover a substantial portion of this large moon's surface. The inset shows a lake near Titan's north pole. It is a bit larger than Lake Superior, one of Earth's largest lakes.

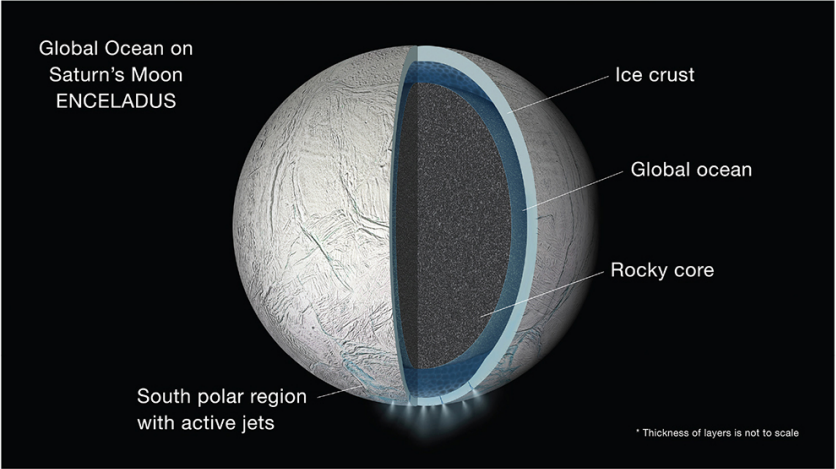


Figure A Cutaway illustration of Enceladus, depicting an icy crust hiding a liquid-water ocean sitting above its rocky core. Layers are not to scale.

Ocean means
endless life
possibilities



Overview of Origin Theories

Panspermia Theory

Life may have originated from space, carried by meteorites or cosmic dust containing microorganisms or **chemical precursors**.

Primordial Soup Theory

Life began in an organic molecule 'soup' on early Earth, energized by lightning and UV radiation to trigger **chemical reactions**.

Hydrothermal Vent Theory

Life may have started near deep-sea hydrothermal vents where mineral-rich water and heat supported **chemical evolution**.

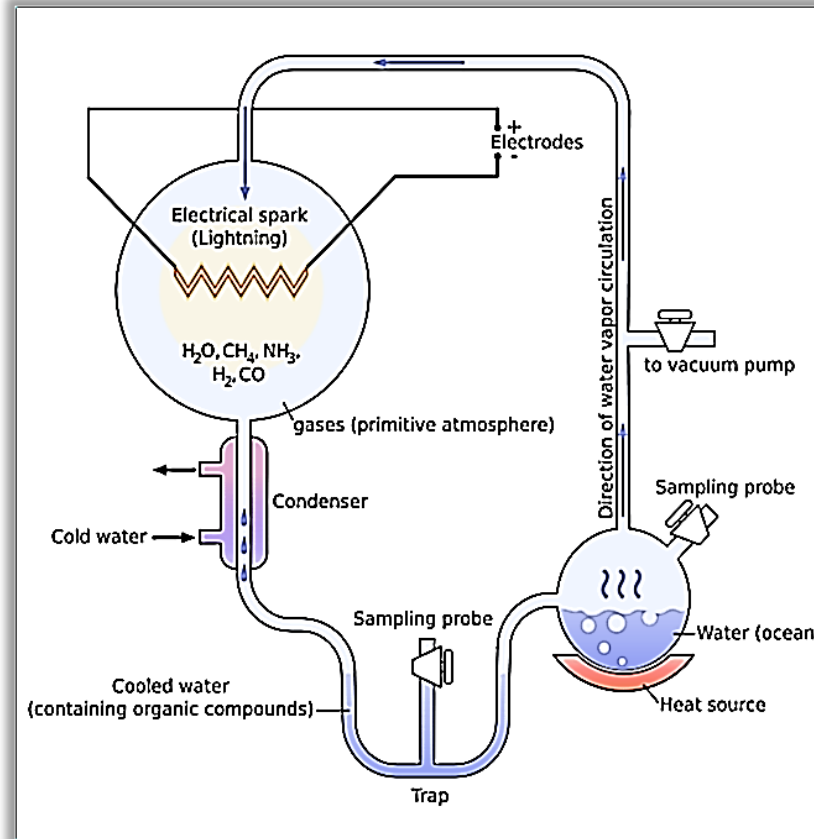
All theories are somehow ocean-related
(life depends on saline-water in cells)

The Miller-Urey Experiment (1953)

Testing the primordial soup theory



Stanley Miller



Early Earth Simulation

Miller recreated early Earth environment using water, methane, ammonia, and hydrogen gases in a closed system.

Energy Source for Reactions

Electric sparks were used to mimic lightning, providing energy to drive chemical reactions forming organic molecules.

Formation of Amino Acids

The experiment produced amino acids, crucial building blocks of life, demonstrating life's chemical origins.

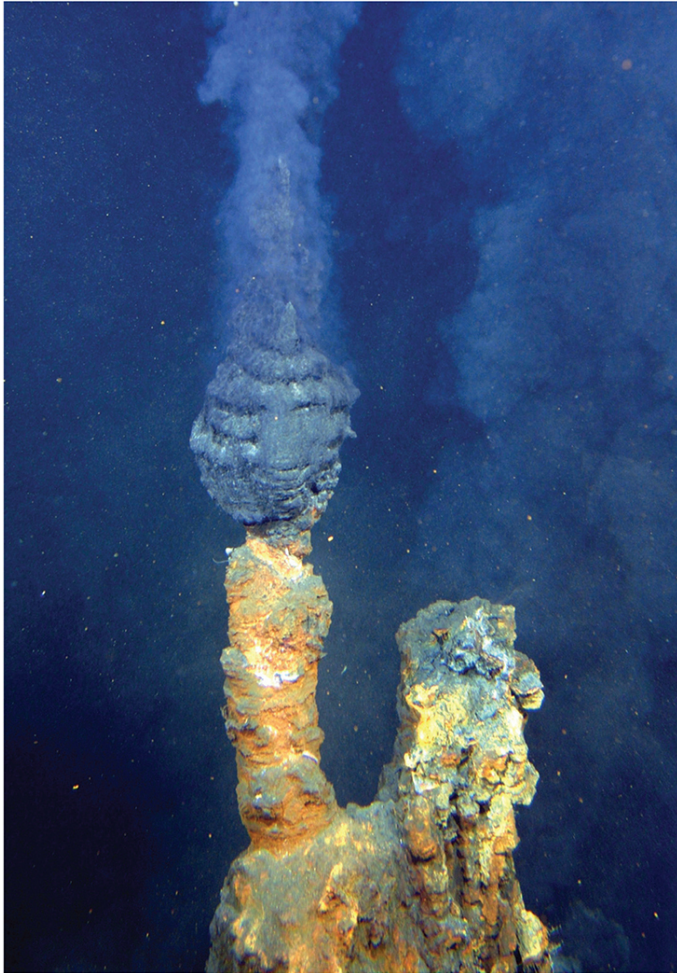


Figure 1.12 An environment for biosynthesis? Weak sunlight and unstable conditions on Earth's surface may have favored the origin of life on mineral surfaces near deep-ocean hydrothermal vents similar to the one shown here. In ancient vents, alkali hydrothermal fluids mixed with more acidic ocean waters to create a proton gradient. This gradient is remarkably similar to the way cells pump protons across their membrane to harness energy.

Image courtesy of New Zealand American Submarine Ring of Fire 2007 Exploration, NOAA Vents Program, the Institute of Geological & Nuclear Sciences and NOAA-OE

Deep Ocean Hydrothermal Vents

Chemical Environment of Vents

Hydrothermal vents emit mineral-rich heated water, creating a unique chemical environment for organic molecule formation.

Protection from Harsh Surface Conditions

Vents offer shelter from early Earth's harsh surface conditions like UV radiation and meteor impacts.

Extremophiles and Life Origin

Extremophiles thriving in vents support the idea that life could have originated in these environments.

Another thinking moment

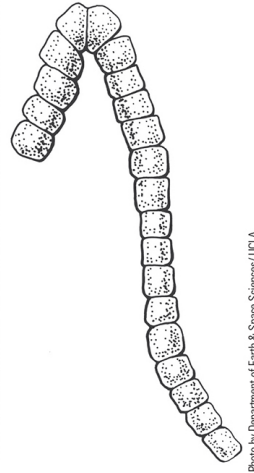
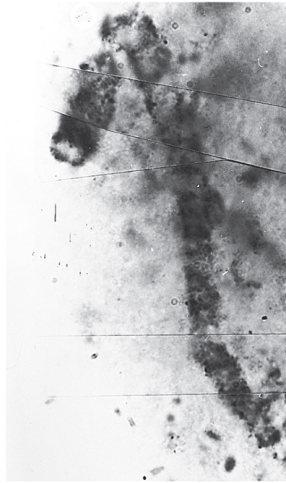
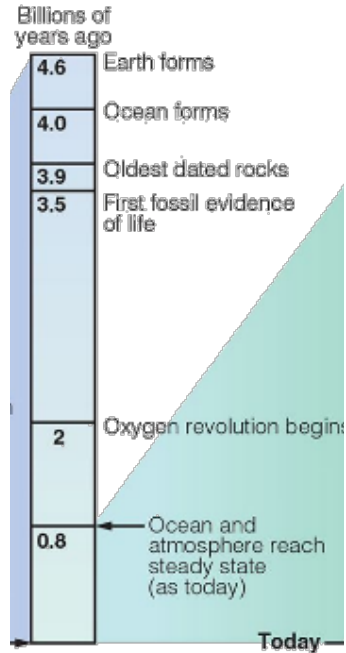


Photo by Department of Earth & Space Sciences / UCLA

Figure 1.13 Fossil of a bacteria-like organism (with an artist's reconstruction) that photosynthesized and released oxygen into the atmosphere. Among the oldest fossils ever discovered, this microscopic filament from northwestern Australia is about 3.5 billion years old.

The oldest fossils of early life forms are of photosynthetic bacteria-like organisms of about 3.5 billion years ago (0.5 billion years later than ocean formation).

What is the inconsistency between such fossil records and our theories about life formation?

- ☐ Bacteria are composed of organic matter like amino acids and proteins
- ☐ Photosynthetic bacteria are the simplest life forms and should have appeared first
- ☐ Theories suggest simpler, and probably non-photosynthetic, organisms appeared first
- ☐ There are no inconsistencies', just a lack of scientific evidence bridging the gap between the theories and the fossils

